REPRODUCING AND EXTENDING PISSA-LORA FOR VIT FINE-TUNING ON CIFAR-10

A reproduction-and-extension study of principal singular value initialization for adapter-based vision transfer.

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Graduate Project

REPO

github

/pissa-lora-vit

FOUR METHODS, ONE FAIR COMPARISON.

The core question is not PiSSA vs. full fine-tuning — it is PiSSA vs. standard LoRA under matched adapter settings.

01

LINEAR PROBE

Freeze the entire ViT encoder. Train only the 10-way classifier head.

TRAINABLE SHARE

0.009%

02

FULL FINE-TUNE

All ViT parameters are trainable. The expensive baseline that adapters aim to replace.

TRAINABLE SHARE

100.000%

03

LORA

Rank-r adapters on attention query and value projections. Backbone frozen.

TRAINABLE SHARE

0.351%

04

PISSA-LORA

Same adapter as LoRA. Initialized from the principal singular components of W.

TRAINABLE SHARE

0.351%

SAME LOW-RANK ADAPTER, SMARTER STARTING POINT.

LoRA learns a low-rank update on top of a frozen weight. PiSSA keeps that structure but seeds it from the principal singular components of W.

01 · STANDARD LORA

Hu et al. 2021

Freeze W. Add a low-rank update.

$$W_{eff} = W + \Delta W$$

$$\Delta W = B \cdot A$$

$$B \in \mathbb{R}^{d \times r},\ A \in \mathbb{R}^{r \times k},\ r \ll \min(d, k)$$

→ Init: A Gaussian, $B = 0 \rightarrow \Delta W(0) = 0$.

→ Effective weight starts at W. The adapter learns the task update from random low-rank directions.

$\Delta W(0) = 0$

02 · PISSA-LORA

Meng et al. 2024

Same adapter. Initialize from the SVD of W.

$$W = U\,\Sigma\,V^T = W_{pri} + W_{res}$$

$$W_{pri} = U_r\,\Sigma_r\,V_r^T \cdot W_{res} = U_{tail}\,\Sigma_{tail}\,V_{tail}^T$$

$$B_0 = U_r\,\Sigma_r^{1/2}\ ,\ A_0 = \Sigma_r^{1/2}\,V_r^T$$

$$W_{base} \leftarrow W_{res}$$

→ $B_0 \cdot A_0 = W_{pri}$, so the adapter already holds the top-r singular part of W.

→ Frozen base is replaced by W_{res} so the effective weight at step 0 is still W.

$W_{eff}(0) = W_{res} + B_0 A_0 = W$

BOTH START FROM THE SAME EFFECTIVE WEIGHT — BUT PISSA POINTS AT THE MOST IMPORTANT SINGULAR DIRECTIONS INSTEAD OF RANDOM LOW-RANK NOISE.

DOES IT EVEN WORK ON VIT?

A small-scale sanity check on Colab. The notebook labels its baseline 'classifier-only', but the freeze block is commented out — that run was actually full fine-tuning.

CONFIGURATION

data split	5k train · 1k val
epochs	1
optimizer	AdamW
learning rate	2e-4
weight decay	0.01
batch size	32
PiSSA rank · α	8 · 16
target modules	query, value

VALIDATION ACCURACY · START → END

FULL FT

0.068

↓

0.920

PISSA-LORA

0.100

↓

0.963

LESSON

PISSA-LORA LOOKED STRONGER THAN FULL FT EVEN IN THIS TINY EARLY RUN — ENOUGH SIGNAL TO INVEST IN A REAL COMPARISON.

ADD THE MISSING LORA BASELINE.

Move from notebook to a structured pipeline on a Tesla T4. Now PiSSA is being tested against standard LoRA, not just Full FT.

SETUP

data	5k train · 1k val
epochs	1
optimizer	AdamW
lr · Full FT	5e-5
lr · LoRA / PiSSA	2e-4
batch · grad accum	16 · 2 (eff. 32)
rank · α · dropout	8 · 16 · 0.05
hardware	Tesla T4

BEST VALIDATION ACCURACY

FULL FT	94.5%
LORA	96.2%
PISSA-LORA	96.4%

Now the comparison is fair. PiSSA edges LoRA by a small but real margin under matched adapter settings.

SCALE UP THE DATA, THE GAP STAYS.

Push to the full 50k training set. PiSSA's lead over LoRA is small but consistent — and it grows a little with more epochs.

SUITE A · 1 EPOCH — 50k train · 10k val

LORA	98.22%
PISSA-LORA	98.34%

GAIN +0.12 pp

SUITE B · 3 EPOCHS — 50k train

LORA	98.38%
PISSA-LORA	98.54%

GAIN +0.16 pp

THE STORY SHARPENS — PISSA MAY NOT WIN BY A HUGE MARGIN, BUT IT TENDS TO BEAT LORA EVERY SINGLE TIME.

REPRODUCIBLE FINAL SETUP.

Multiple seeds, cleaner configs, cluster GPUs. The point is no longer to prove it works — it is to measure how well, with proper variance.

01 · ENVIRONMENT

HARDWARE
Purdue RCAC Gautschi · NVIDIA H100

BACKBONE
ViT-Base patch16-224 · in21k

DATASET
CIFAR-10

SEEDS
7 · 42 · 123

02 · OPTIMIZATION

optimizer	AdamW
weight decay	0.01
epochs	3
lr · Full FT	5e-5
lr · Linear Probe	2e-4
lr · LoRA / PiSSA	2e-4

03 · PROTOCOL

TRAIN AUGMENTATION
Random resized crop · horizontal flip

EVAL TRANSFORM
Resize · center crop

REPORTED METRICS
Mean best val acc · std · trainable params · time

THREE SUITES — QUICK ABLATION, DATA-REGIME SWEEP, RANK ABLATION — SAME BACKBONE, SAME PROTOCOL, THREE SEEDS EACH.

PISSA-LORA BEATS FULL FT WITH 284× FEWER PARAMETERS.

5k train · 2k val · 3 epochs · 3 seeds. Mean best validation accuracy reported with standard deviation.

METHOD	TRAINABLE %	MEAN BEST ACC	TRAIN TIME
LINEAR PROBE	0.009	94.58 ± 0.31	7.87s
FULL FT	100.000	97.12 ± 0.33	19.38s
LORA	0.351	97.22 ± 0.26	20.94s
PISSA-LORA	0.351	97.53 ± 0.33	21.15s

TRAINABLE PARAMETERS

FULL FT	85,806,346
PISSA	302,602

HEADLINE RATIO

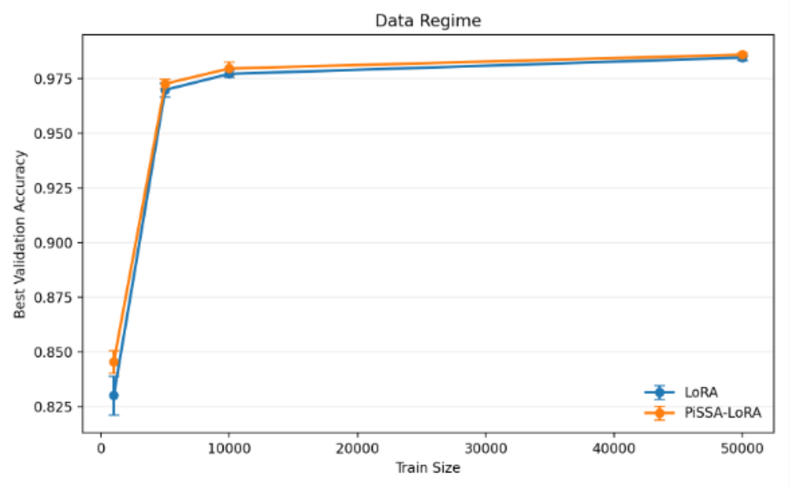
284×

fewer trainable parameters than Full FT, while reaching higher mean accuracy. Only 0.351% of the model is updated.

PISSA HELPS MOST WHEN DATA IS SCARCE, PEAKS AT RANK 16.

DATA REGIME SWEEP

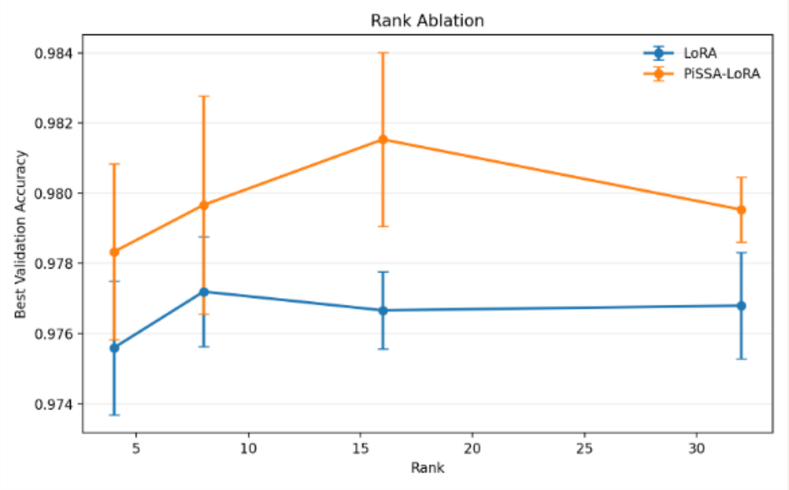
1k → 50k train



1K	5K	10K	50K
+1.53 pp	+0.28 pp	+0.25 pp	+0.13 pp

RANK ABLATION

r ∈ {4, 8, 16, 32}



BEST RANK · 16

LORA	97.67 ± 0.11
PISSA	98.15 ± 0.25

READING THE TREND

PISSA WINS AT EVERY DATA SCALE, BUT THE LIFT IS LARGEST IN THE LOW-DATA REGIME WHERE GOOD INITIALIZATION MATTERS MOST.

ON RANK

PISSA BEATS LORA AT ALL RANKS. DOUBLING RANK PAST 16 STOPS PAYING FOR ITSELF.

A RELIABLE IMPROVEMENT, NOT A REVOLUTION.

01 **PISSA-LORA CONSISTENTLY BEAT STANDARD LORA.**

Across every quick ablation, data regime, and rank tested in this study.

02 **IN THE 5K QUICK ABLATION IT SLIGHTLY SURPASSED FULL FT.**

97.53% vs. 97.12% mean best val acc — with 284× fewer trainable parameters.

03 **THE BIGGEST GAIN SHOWS UP IN THE LOW-DATA REGIME.**

+1.53 pp at 1k images, narrowing to +0.13 pp by 50k as optimization gets easier.

04 **THE WIN IS PARAMETER EFFICIENCY AND ACCURACY, NOT RAW SPEED.**

Adapter methods are roughly comparable in wall-clock to Full FT on H100 at this scale.

TAKEAWAY

PISSA-STYLE INITIALIZATION IS A PRACTICAL IMPROVEMENT FOR VIT ADAPTER TUNING ON CIFAR-10 — MOST USEFUL WHEN DATA IS LIMITED.

THANK YOU.

Suggestions and questions are very welcome.

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PROJECT REPOSITORY

github.com

/pissa-lora-vit